

Integrated AV Development Suite

Complete User Guide

Version 1.0 - February 2026

Overview & Purpose

The Integrated AV Development Suite bridges real-world infrastructure and virtual testing. It enables autonomous vehicle developers to plan routes using actual UK charging infrastructure, then automatically generate realistic simulation scenarios to test AV systems under various conditions.

Tab 1: BigQuery Setup

Purpose: Connect to Google Cloud BigQuery with 42,000+ UK EV charging points.

Process: Upload JSON key → Test connection → Load charging point data

Background: OAuth authentication, SQL query execution, spatial data conversion to CRS 4326

AV Application: Validates charging infrastructure coverage before AV fleet deployment.

Tab 2: Road Network

Purpose: Download OpenStreetMap road network and convert to routing graph.

Process: Enter location → Download OSM data → Create dodgr graph

Background: OSM Overpass API query, filters motorway/trunk/primary/secondary roads, creates weighted graph (~35,000 edges for Cambridge)

AV Application: Provides ground-truth road topology for AV path planning validation.

Tab 3: Route Optimizer

Purpose: Calculate optimal route through nearest charging stations.

Process: Select origin/destination → Set charging points (1-10) → Calculate → Send to Simulator

Background: Geocoding with tmaptools, spatial distance calculation, Dijkstra shortest path on road graph, route optimization across multiple charging points

AV Application: Optimizes AV fleet routing to minimize range anxiety. Critical for autonomous taxi services.

Tab 4: Route Map

Purpose: Visualize calculated route on interactive map with actual road paths.

Process: View route automatically after calculation

Background: `dodgr_flows_aggregate()` extracts actual road edges used in route, draws each edge as polyline following real roads (not straight lines)

AV Application: Validates planned routes follow actual drivable roads.

Tab 5: Omniverse Connection

Purpose: Connect to NVIDIA Isaac Sim and generate AV test scenarios.

Process: Test connection → Enter cities → Configure weather/traffic/road type → Generate scenarios

Background: Flask API HTTP POST to /scenarios/query, Isaac Sim launches Kia Niro EV model, generates camera data (lane detection + sign recognition), LiDAR point clouds, GPS trajectories, incident detection, quality scoring (1-10)

AV Application: Generates synthetic training data for AV perception systems. Tests corner cases (fog, night) that are dangerous in real world.

Tab 6: Scenarios

Purpose: Browse, filter, and analyze generated simulation scenarios.

Features: Interactive DataTable with sort/search/filter, AV Readiness (GREEN 8-10 production ready, AMBER 5-7 needs improvement, RED 1-4 unsafe), Quality scoring based on road geometry/traffic/weather/sensors, Trajectory plots, Incident analysis (hard braking, lane departure, near miss, sensor failure)

AV Application: Data-driven development decisions. Compare scenarios to identify failure conditions, prioritize improvements.

Tab 7: Route Preview

Purpose: Geographic visualization of scenario quality and incidents on map.

Features: Quality markers (green=safe, orange=caution, red=unsafe), Incident markers with details, Route path colored by quality, Weather/traffic impact visualization

AV Application: Geographic risk assessment. Identify specific intersections/segments that reduce AV safety.

Tab 8: Advanced Visualization

Purpose: Advanced AR/VR capabilities for immersive scenario exploration.

AV Application: Stakeholder communication and training through immersive visualization.

Complete Integration Workflow

Phase 1 - Real World: Connect BigQuery → Download network → Calculate route → View map

Phase 2 - Export: Click 'Send to Simulator' - converts route to scenario with GPS waypoints and conditions

Phase 3 - Virtual Test: Configure conditions → Generate scenarios → Analyze results → Review map

Phase 4 - Decisions: Identify RED zones → Compare conditions → Iterate with variations

Industry Applications

Robotaxi Services	Route safety validation, weather condition assessment
Autonomous Delivery	Warehouse-to-customer optimization with charging
Public Transport	AV bus route planning with reliable scheduling
Logistics	Long-haul truck routes with mandatory charging stops
AV Development	Synthetic training data, edge case testing
Urban Planning	Infrastructure gap analysis, charging station placement

Technical Benefits

- **Risk Reduction:** Test dangerous scenarios virtually before real deployment
- **Cost Savings:** 99% reduction vs real-world testing (£10 vs £10,000+ per test)
- **Scalability:** Generate thousands of scenarios in hours
- **Reproducibility:** Exact conditions can be re-tested
- **Regulatory Compliance:** Document AV performance for safety submissions
- **Continuous Validation:** Re-test after software updates

Summary

The Integrated AV Development Suite provides comprehensive workflow from real-world route planning to virtual safety validation. By combining actual UK charging infrastructure (42,000+ points) with photorealistic NVIDIA Isaac Sim, developers identify and fix AV safety issues before deployment, significantly reducing risk and costs. The 'Send to Simulator' integration ensures simulation testing reflects actual operational routes.

Key Metrics:

- 8 integrated modules (4 route planning + 4 simulation)
- 42,682 UK charging points from BigQuery
- ~35,000 road edges per network (Cambridge example)
- Quality scoring: 1-10 scale with GREEN/AMBER/RED classification
- Sensor data: Camera (lanes + signs) + LiDAR (obstacles)
- Incident types: Hard braking, lane departure, near miss, sensor failure